

Q6
DECLARE vehicle_type, membership, disabled_priority, station_available = STRING
DECLARE current_time = INTEGER
DECLARE battery_level, required_level, required_charging = REAL
DECLARE parking_duration = REAL
DECLARE charging_priority, peak_status, warning_message = STRING
DECLARE charging_rate, charging_cost = REAL
DECLARE parking_cost, parking_discount = REAL
DECLARE final_amount = REAL

charging_rate $\leftarrow$ 0, charging_cost $\leftarrow$ 0, parking_cost $\leftarrow$ 0
parking_discount $\leftarrow$ 0, final_amount $\leftarrow$ 0, required_charging $\leftarrow$ 0
charging_priority $\leftarrow$ " ", peak_status $\leftarrow$ " ", warning_message $\leftarrow$ " "

PRINT "Enter vehicle type (E=Electric, H=Hybrid) ="

INPUT vehicle_type

PRINT "Enter current battery level (%):"

INPUT battery_level

PRINT "Enter required charging level (%):"

INPUT required_level

PRINT "Enter expected parking duration ="

INPUT parking_duration

PRINT "Enter current time ="

INPUT current_time

PRINT "Does the customer have a parking membership? (Y/N) ="

INPUT membership

PRINT "Does the customer have disabled person priority status? (Y/N) ="

INPUT disabled_priority

PRINT "Is the charging station available? (Y/N):"
 INPUT station-available
 PRINT "Vehicle type:", vehicle-type
 PRINT "Current Battery Percentage:", battery-level

IF station-available = N THEN

IF vehicle-type = H THEN

PRINT "Charging unavailable - Parking only"

ELSE

PRINT "No charging slot available"

ENDIF

ELSE

IF vehicle-type = E THEN

charging-priority $\leftarrow$ Eligible for charging

ELSE IF vehicle-type = H THEN

IF battery-level $<$ 40 THEN

charging-priority $\leftarrow$ Eligible for charging

ELSE

PRINT "Vehicle does not qualify for EV charging"

charging-priority $\leftarrow$ Parking only

ENDIF

ENDIF

required-charging $\leftarrow$ required-level - battery-level

PRINT "Required Charging Percentage:", required-charging

IF required-charging $\leq$ 0 THEN

PRINT "No charging required"

charging-priority $\leftarrow$ No charging required

charging-cost $\leftarrow 0$

ELSE IF charging priority = Eligible for charging THEN

IF battery level $C=15$ AND required level ≥ 80 THEN

charging priority $\leftarrow$ emergency charging priority

ELSE IF disabled priority = Y OR membership = Y

AND battery level $C=30$ THEN

charging priority $\leftarrow$ Priority charging

ELSE

charging priority $\leftarrow$ Normal charging

ENDIF

IF current time < 17 OR current time > 22 THEN

peak status $\leftarrow$ off-peak

charging rate $\leftarrow 35$

ELSE

peak status $\leftarrow$ Peak

charging rate $\leftarrow 50$

ENDIF

charging cost $\leftarrow$ required charging $\times$ charging rate

IF charging priority = Emergency charging priority THEN

charging cost $\leftarrow$ charging cost

ELSE IF peak status = Peak THEN

charging cost $\leftarrow$ charging cost $-$ (charging cost $\times 10/100$)

ELSE IF membership = Y THEN

charging cost $\leftarrow$ charging cost $-$ (charging cost $\times 20/100$)

ENDIF

ENDIF

ENDIF

IF parking duration ≤ 2 THEN

parking cost $\leftarrow 200$

ELSE IF parking duration ≤ 5 THEN

parking cost $\leftarrow 400$

ELSE

parking cost $\leftarrow 700$

ENDIF

IF disabled priority = Y THEN

parking discount $\leftarrow$ parking cost

parking cost $\leftarrow 0$

ELSE IF membership = Y THEN

parking discount $\leftarrow$ parking cost $\times 20/100$

parking cost $\leftarrow$ parking cost - parking discount

ELSE

parking discount $\leftarrow 0$

ENDIF

final amount $\leftarrow$ charging cost + parking cost

IF parking duration > 8 THEN

warning message $\leftarrow$ long stay warning - Please rebote your vehicle after charging

ELSE

warning message $\leftarrow$ Standard parking duration

ENDIF

PRINT "Vehicle type:", vehicle type

PRINT "Current Battery Percentage", battery level

PRINT "Required Charging Percentage:", required charging

PRINT "Charged Priority:", charging priority

PRINT "Peak/Off-Peak season", peak-status

Date: _____

```
PRINT "Charging cost:Rs.", charging cost  
PRINT "Parking cost:Rs.", parking cost  
PRINT "Parking Discount:Rs.", parking Discount  
PRINT "Final Payable Amount:Rs", final amount  
PRINT "Warning message:" warning message  
END
```